Chapter 04

AI Governance

Why AI Governance Matters?

Rapid adoption of AI tools and agents

Increased legal and compliance exposure

Legal and Compliance Considerations

AI use may be regulated or restricted

Requirements vary by organization and industry

Documenting How AI is Used

Some organizations require disclosure

Identifies AI tools used and for what purpose

AI Usage Restrictions and Polices

AI use may be restricted by industry regulations

Organizations define acceptable AI use through policy

Policies specify approved tools and prohibited uses

Sensitive Data Risks and AI Data Leakage

Sensitive data may be submitted to AI systems

Data can become part of the AI model

Other users or organizations may retrieve it

Chapter 04

AI Risk

Risk of Using AI

AI introduces accuracy, security, and trust risks

Risks must be understood before adoption

Common AI Risks

Hallucinations

Data exposure

Model poisoning

Malicious prompts

AI Hallucinations

AI presents false information as fact

No supporting evidence for the output

Data Exposure Risks and AI Data Leakage

Data submitted to AI may become part of the model

Sensitive information may be exposed to others

Model Poisoning

Attackers manipulate training data or behavior

Produces biased, incorrect, or attacker-controlled results

Malicious Prompts

Inputs designed to bypass AI safeguards

Can leak data or trigger unsafe actions

Chapter 04

AI Use

Common AI Use Cases

Data comparison and analysis

Log file analysis

Document creation

Incident investigation

Automation and orchestration

Data Analysis and Log File Analysis

AI quickly analyzes large data sets

Converts complex logs into human-readable summaries

Document Creation

Provides a starting point for documentation

Assists with routine reports and summaries

Incident Investigation and Event Correlation

Ingests multiple data sources

Identifies suspicious or malicious activity across systems

Automation and Orchestration

Automatically responds to incidents

Executes predefined response actions and scripts

Chapter 04

Automation - Integration

Automation and APIs

Automation increases with complexity

APIs enable system interaction

Interface for predefined actions

AI Concept Example

Structured interface for tasks

Limited predefined operations

Not full system access

APIs in Modern Platforms

Programmable interfaces for applications

Enable automation and integration

Common in web platforms

Webhooks

Automated HyperText Transfer Protocol (HTTP) messages

Notify clients of changes

Browser callback behavior

The Polling Model

Repeated update requests

Continuous change checking

High overhead

Webhooks vs Polling

Webhooks

- Push updates on change

Polling

- Constant pull requests

Reduced overhead with webhooks

Plugins
Extending Functionality

Add features to tools

Extend base product

Not included by default

Plugin Integration and Licensing

Use APIs for interaction

Free or pain options

Separate licensing possible

Chapter 04

Automation - Scripting

Shell Scripting Overview

Oldest scripting type

Built into Linux and Unix

Executes ordered or conditional commands

Supports branching based on input

Identifying a Shell Script

Begins with a shebang (#!)

Points to interpreter

- Example: /bin/bash

Variables often uppercase

```
#!/bin/bash
echo "Starting backup"
tar -czf backup.tar.gz /home/user
```

Python Scripting Characteristicis

Widely used in organizations

Easy to read syntax

Large library ecosystem

No semicolons

Python Formatting and Readability

Uses whitespace formatting

Structure based on indentation

Reduces syntax errors

Improves readability

```
log_entries = ["login success", "login failure"]
for entry in log_entries:
    print(entry)
```

PowerShell Overview

Built into Windows

Command-line automation tool

Similar to shell scripting

Used for system management

```
Get-Content security.log
Add-ADUser -Name "jsmith"
Set-ADUser -Enabled $true
```

PowerShell Verb-Noun Syntax

Commands use verb-noun format

- Verb
  - Get, add, set
- Noun
  - User, content, group

Example commands

- get-content
- add-user

Identifies PowerShell scripts

Chapter 04

Response Standards

Consistency in Incident Response

Consistency ensures secure systems

Prevents gasps in protection

Standardizes response across teams

Ensures incidents are handled correctly

Standardized Response Processes

Standard process guide actions

Enable coordination across teams

Apply regardless of responder

Ensure consistent outcomes

Standard Operating Procedures (SOPs)

Define how situations are handled

Provide structured response steps

Support team coordination

Ensure consistent execution

Guardrails in Reponse

Guide employee actions

Allow flexibility in execution

Define boundaries for decisions

Balance control and autonomy

Playbooks vs Runbooks

Playbooks

- Team response procedures

Runbooks

- Individual task guidance

Both provide
step-by-step actions

Support coordinated response